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Battery Cathodes

Tuning Solvation Structure Via Inductive and Steric Hindrance Effects for High-Voltage $LiCoO_2$ Batteries

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Abstract: High-voltage lithium cobalt oxide ($LiCoO_2$) has a high specific capacity and energy density, making it a promising cathode material for next-generation lithium-ion batteries (LIBs). However, stabilizing $LiCoO_2$ at elevated charging cut-off voltages remains challenging due to the severe interfacial degradation, particularly the instability of the cathode–electrolyte interphase (CEI) under oxidative conditions. Herein, we propose a novel push–pull electrolyte design strategy by incorporating a non-coordinating diluent, 1H,1H,5H-perfluoropentyl-1,1,2,2-tetrafluoroethylether (HFE), which modulates the Li^+ solvation structure through strong inductive and steric hindrance effects, thereby enabling the formation of a robust CEI on the $LiCoO_2$ surface. Guided by molecular electrostatic potential analysis and nuclear magnetic resonance characterizations, the optimized electrolyte creates a tailored solvation environment that suppresses parasitic interfacial reactions and facilitates the formation of a bilayer interphase. As a result, the $LiCoO_2$ cathode exhibits excellent electrochemical stability with the HFE-containing electrolyte, delivering over 300 stable cycles at 4.6 V in high-loading $LiCoO_2$//Li cells (~11 mg cm^{-2}) and retaining 77% capacity after 200 cycles in $LiCoO_2$//Graphite full-cells at 4.5 V. This solvation engineering strategy provides a promising pathway toward next-generation high-voltage $LiCoO_2$-based batteries.

Introduction

Lithium cobalt oxide ($LiCoO_2$), owing to its high theoretical capacity (274 mAh g^{-1}) and excellent cycling stability, has long served as a benchmark cathode material for lithium-ion batteries (LIBs), particularly in consumer and portable electronics.[1–5] To meet the growing demands of 3C electronics and 5G smart devices, it is highly desirable to improve the energy density of $LiCoO_2$ by increasing their charging cut-off voltage (above 4.3 V).[6–9] However, increasing the charging cut-off voltage accelerates electrolyte decomposition due to the highly oxidative environment of delithiated Li_xCoO_2 and the insufficient oxidative stability of conventional carbonate electrolytes,[10–13] which together lead to rapid performance degradation.[14,15]

Liquid electrolyte engineering has emerged as a promising strategy to resolve the interfacial incompatibility between high-voltage cathodes and conventional electrolytes.[16–19] Recent efforts have focused on regulating Li^+ solvation structure to improve electrolyte stability and facilitate the formation of robust cathode–electrolyte interphase (CEI), thereby enhancing the overall electrochemical performance. Representative approaches include high-concentration electrolytes (HCEs)[20,21] and localized high-concentration electrolytes (LHCEs)[22,23] (Figure 1). These electrolytes promote anion-rich solvation environments, wherein anions coordinate directly with Li^+ to form contact ion pairs (CIPs) and ion aggregates (AGGs), ultimately contributing to stable anion-derived interphases.[24–27] However, the large quantities of lithium salt and inert diluent required by HCEs and LHCEs significantly increase cost, limiting their practicality for large-scale applications. To overcome this limitation, weakly solvating electrolytes (WSEs)[28–31] have emerged as a promising strategy to enhance the high-voltage performance of $LiCoO_2$-based batteries. By employing weakly coordinating solvents such as fluoroethylene carbonate (FEC) and methyl trifluoroethyl carbonate (FEMC), the strength of Li^+–solvent coordination is significantly reduced.[32,33] As a result, the solvation structure shifts from solvent-separated ion pairs (SSIPs) toward CIPs and AGGs, thereby promoting anion-derived interfacial chemistry. However, typical fluorinated electrolyte systems, such as 1.0 M $LiPF_6$ in FEC/FEMC (1:1 by volume, denoted as FF), show insufficient performance under high-voltage cycling, highlighting the need for further regulation of the electrolytes with an appropriate solvation structure.[34]

Herein, we propose an electrolyte design strategy that introduces a fluorinated ether diluent, 1*H*,1*H*,5*H*-perfluoropentyl-1,1,2,2-tetrafluoroethylether (HFE), into a

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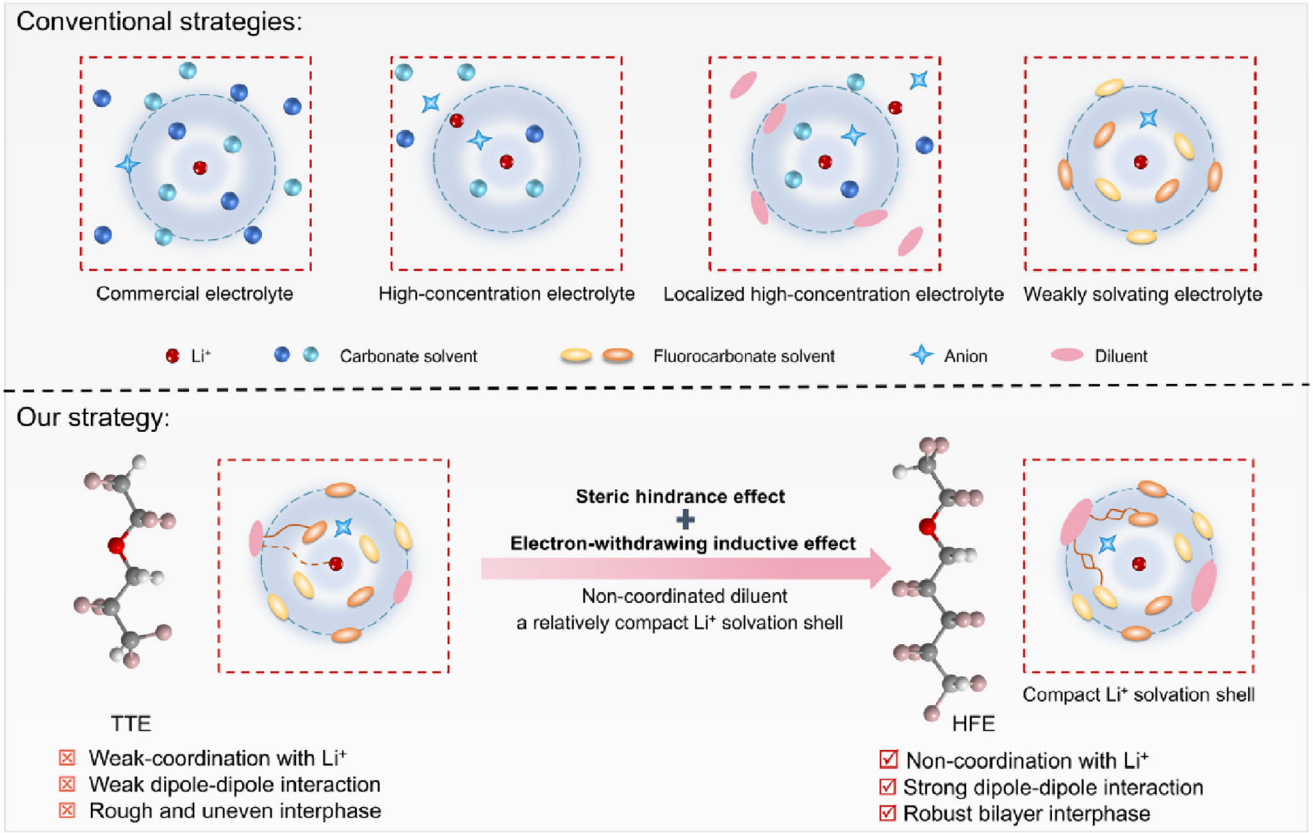


Figure 1. Schematic diagram of the solvation structures in conventional electrolytes and our electrolyte design strategy.

conventional all-fluorinated electrolyte (denoted as FFHB) to enhance the electrochemical stability of $LiCoO_2$ batteries at high cut-off voltages (>4.5 V). Owing to its high steric hindrance and strong electron-withdrawing inductive effects, HFE effectively weakens Li^+–solvent interactions and promotes the formation of anion-rich solvation sheath. This tailored solvation environment facilitates the formation of a robust bilayer CEI, composed of an inorganic-rich inner layer (LiF, Li_2O) and a mechanically flexible, boron-rich organic outer layer. As a result, the optimized FFHB electrolyte exhibits outstanding high-voltage performance, enabling stable cycling of high-loading $LiCoO_2$//Li cells (~11 mg cm^{-2}) for over 300 cycles at 4.6 V, and achieving 77% capacity retention after 200 cycles in $LiCoO_2$//Graphite full-cells at 4.5 V. This work offers a practical and scalable pathway toward next-generation high-voltage $LiCoO_2$ batteries through solvation structure engineering.

Results

Solvation Structure Analysis of the Electrolytes

To elucidate the interaction between solvents and Li^+, we employed the electrostatic potential (ESP) descriptor to evaluate their nucleophilicity and coordination tendencies with Li^+.[35–37] Among the solvents studied, FEC exhibits the most negative ESP_{min}, indicating a strong electron-donating ability and a high affinity for Li^+ coordination (Figure 2a). In contrast, 1,1,2,2-tetrafluoroethyl-2,2,3,3-tetrafluoropropylether (TTE) and HFE show much less negative ESP_{min} values, suggesting that they are poor electron donors and exhibit weak interactions with Li^+.[23] Notably, both HFE and TTE contain tetrafluoroethyl group (—C_2F_4H), which exhibits pronounced inductive effect due to its strong electron-withdrawing property. Compared to the —H group in TTE, 1*H*,1*H*,5*H*-perfluoropentyl group in HFE provides an additional —C_2F_4H moiety (Figure S1), exhibiting a significantly stronger electron-withdrawing inductive effect ($-I$ effect: —C_2F_4H > —H). This inductive withdrawal results in a more uniform dispersion of the negative charge across the oxygen and fluorine atoms, further diminishing their ability to coordinate with Li^+.[35] To gain a deeper insight into the Li^+ coordination behavior of the fluorinated diluents, we conducted ^{19}F nuclear magnetic resonance (NMR) spectroscopy to understand Li^+ coordinating interactions with TTE and HFE in the all-fluorinated electrolytes FFTB (1 M $LiPF_6$ and 0.02 M LiDFOB in FEC/FEMC/TTE, 3:5:2 by volume) and FFHB (same salt ratio in FEC/FEMC/HFE, 3:5:2 by volume). ^{19}F NMR spectrum reveals a notable difference in the upfield shift of fluorine atoms in the presence of Li salts, particularly in the TTE-containing FFTB electrolyte (Figure 2b). The fluorine atoms in the 2,2,3,3-tetrafluoropropyl group of TTE exhibit pronounced upfield shifts of 0.21 and 0.18 ppm, while those in the 1,1,2,2-tetrafluoroethyl group show smaller shifts of 0.12 and 0.10 ppm, respectively. This difference

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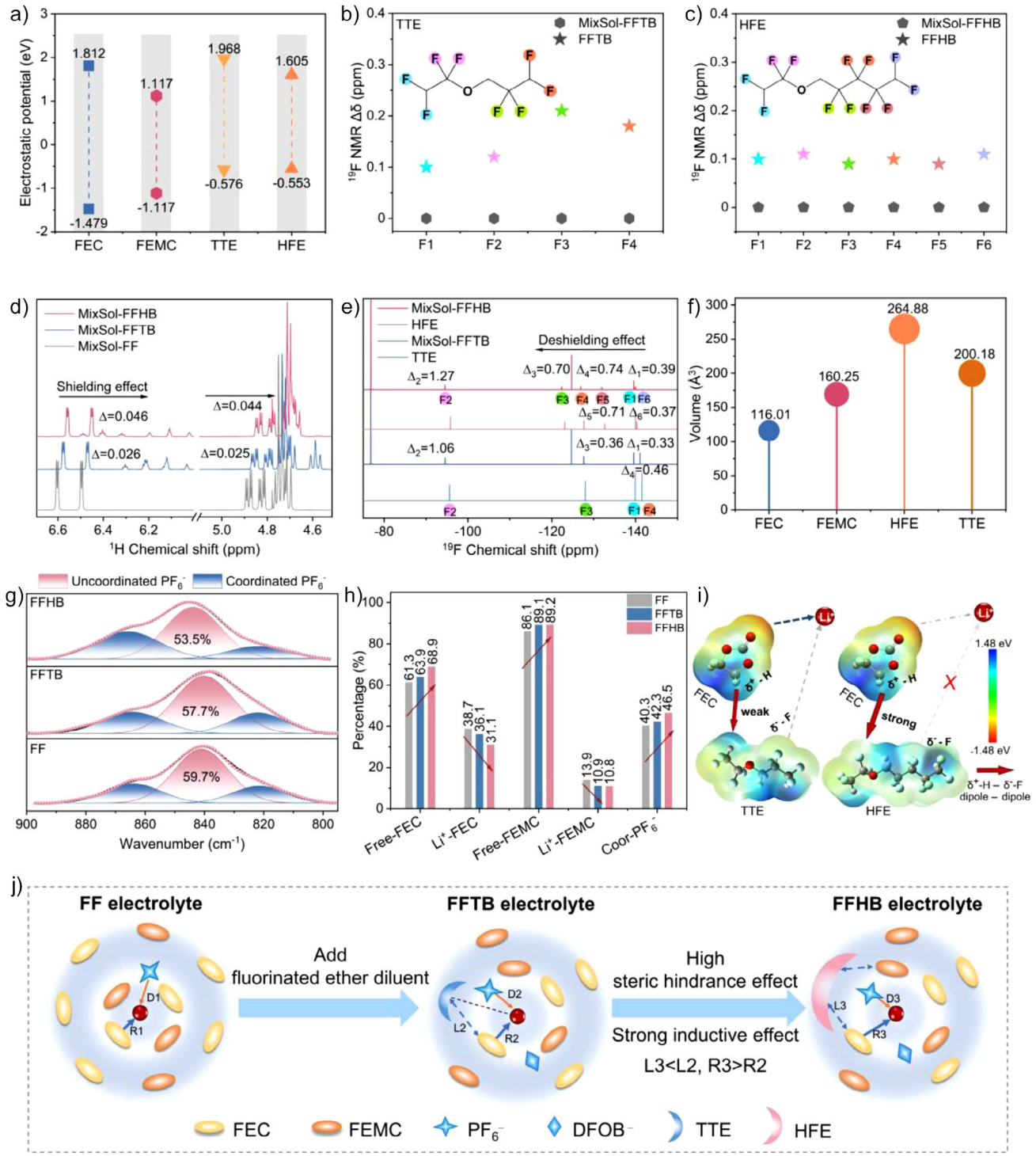


Figure 2. The design rationale and characterization of high-voltage electrolyte. a) ESP of various solvents. ^{19}F NMR chemical shift variations of b) FFTB and c) FFHB electrolytes before and after lithium salt addition. d) ^{1}H NMR spectra of FEC in three mixed solvents. e) ^{19}F NMR spectra of mixed solvents and diluents. f) Molecular volumes of various organic solvents. g) FTIR spectra of FF, FFTB, and FFHB electrolytes. h) Proportion of solvated and free solvents. i) Schematic illustration of the interactions between Li^+, FEC, and different diluents in the electrolytes. j) Illustration of solvation structure changes in FF, FFTB, and FFHB electrolytes.

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indicates a stronger coordination interaction between Li^+ and the fluorine atoms in the 2,2,3,3-tetrafluoropropyl group of TTE.[38,39] In contrast, all fluorine atoms in HFE display only slight chemical shifts changes (0.10 and 0.11 ppm for 1,1,2,2-tetrafluoroethyl group, 0.09, 0.10, 0.09, and 0.11 ppm for 1*H*,1*H*,5*H*-perfluoropentyl group, respectively), which can be attributed to the strong $-I$ effect of the $-C_2F_4H$ group, suggesting negligible direct interaction between HFE and Li^+ (Figure 2c). It should be noted that the slight chemical shift changes observed in both HFE and the 1,1,2,2-tetrafluoroethyl group of TTE are likely attributed to push–pull competitive relationship between Li^+–solvent–diluent.[40] These shifts arise from intermolecular interactions between Li^+ and FEC solvent, which subsequently influence the electronic environment of diluents through solvent–diluent interactions.[41,42]

Besides the ion–dipole (ion–solvent) interactions that affect the solvation structure, dipole–dipole (solvent–solvent) interactions also play a significant role in modulating the solvation environment.[42–44] To investigate dipole–dipole interactions, the solvation structure of electrolytes was further analyzed by ^1H NMR and ^{19}F NMR. We systematically compared three distinct mixed solvent formulations of the electrolytes, denoted as MixSol-FF, MixSol-FFTB, and MixSol-FFHB. As shown in Figure 2d, the ^1H NMR spectra of FEC reveal progressively larger upfield shifts in the order of MixSol-FF < MixSol-FFTB < MixSol-FFHB. Compared to the MixSol-FF system, upfield shifts of 0.026 and 0.025 ppm for FEC were observed in the MixSol-FFTB mixture upon addition of diluents, whereas significantly larger shifts of 0.046 and 0.044 ppm were detected in the MixSol-FFHB system. This can be attributed to the stronger dipole–dipole interactions between HFE and FEC, which increase electron cloud density around the hydrogen atoms of FEC.[45] This trend is further supported by the ^{19}F NMR analysis (Figure 2e), where all ^{19}F signals of diluents are shifted downfield, indicating a reduced electron cloud density on the fluorine atoms. Notably, compared to MixSol-FFTB, the upfield shifts of hydrogen atoms in FEC and downfield shifts of fluorine atoms in HFE are more pronounced in MixSol-FFHB (Figure S2), suggesting stronger dipole–dipole interactions between FEC and HFE in MixSol-FFHB. This observation is further supported by density functional theory (DFT) calculations, which show that the binding energy of FEC–HFE is higher than that of FEC–TTE (Figures 2i, S3). These enhanced dipole–dipole interactions weaken the ion–dipole interactions between Li^+ and FEC, thereby facilitating the incorporation of PF_6^- into the Li^+ solvation sheath (i.e., L3 < L2, R3 > R2, where L and R represent interaction distances in FEC–diluent and FEC–Li^+, respectively, with longer distances indicating weaker interactions, Figure 2j). Moreover, the bulky structure of HFE contributes additional steric hindrance that promotes Li^+–PF_6^- coordination. As shown in Figure 2f, HFE exhibits significantly higher steric hindrance than TTE. Due to its intrinsically negligible affinity for Li^+, HFE remains outside the primary solvation shell but effectively occupies the surrounding solution space, thereby maximizing compression of the solvation structure and increasing the interaction frequency between Li^+ and PF_6^-. Consequently, PF_6^- is more effectively incorporated into the solvation sheath in FFHB electrolyte compared to FFTB (D3 < D2). The variation in the Li^+–solvent interactions significantly influence anion coordination within the solvation structure, thereby playing a crucial role in constructing high-voltage resistant interphases. To further elucidate the coordination behavior of anions with Li^+, we deconvoluted the stretching bands from Fourier transform infrared (FTIR) spectroscopy in different electrolytes (Figures 2g,h, S4). Quantitative analysis[18,46–49] reveals that the fraction of the solvated FEC (Li^+–FEC) decreased from 38.7% in the FF electrolyte to 36.1% in FFTB and further to 31.1% in FFHB. In contrast, the proportion of coordinated PF_6^- increased from 40.3% in FF to 42.3% in FFTB and reached 46.5% in FFHB (Figure 2g,h). These FTIR results were further corroborated by molecular dynamics (MD) simulations (Figure S5). In FF electrolyte, the coordination number of FEC is 3.02, which is notably higher than that in FFTB (2.30) and FFHB (2.06) electrolytes. Conversely, the coordination number of PF_6^- shows an opposite trend: the radial distribution function (RDF) signal reveals the strongest first-shell peak in the FFHB electrolyte, with a coordination number of 2.25, followed by FFTB (2.00) and FF (1.42) electrolytes. These findings confirm that more PF_6^- enter into the Li^+ solvation shell in the FFHB electrolyte, which was further validated by NMR analysis in different electrolytes (Figures S6, S8). In the FFHB electrolyte, the ^{19}F NMR signal of PF_6^- exhibits a clear upfield shift compared to that in FFTB, indicating stronger Li^+–PF_6^- interactions. This is accompanied by a corresponding upfield shift of the ^7Li signal, suggesting that the enhanced ion pairing leads to greater electronic shielding around Li^+. Raman spectra also demonstrate a blue shift of the PF_6^- band in FFHB, which is consistent with the enhanced Li^+–PF_6^- interactions detected by NMR (Figure S7). Notably, the FFHB with HFE diluent also induces an upfield shift in the ^{19}F NMR of $DFOB^-$, in comparison to FFTB (Figure S8). This suggests a stronger interaction between Li^+ and the carbonyl functional groups of $DFOB^-$ in the FFHB. Such enhanced interactions facilitate the preferential oxidation of $DFOB^-$ at the cathode, thereby promoting the formation of a robust, anion-derived CEI layer.

As summarized in Figure 2j, the differences in solvation structures among FF, FFTB, and FFHB electrolytes are clearly illustrated. By introducing a non-coordinating diluent into conventional all-fluorinated electrolyte system, weak interactions between the non-coordinating diluent and main solvents facilitate more anion participation in the Li^+ solvation sheath. Meanwhile, the steric hindrance imposed by the diluent contributes to the formation of a more compact solvation structure. This tailored solvation environment not only improves the oxidative stability of the electrolyte but also enhances interfacial stability through the formation of an anion-rich CEI. It deserves mention that the high degree of fluorination endows these electrolytes with excellent flame-retardant properties (Figure S9). Together, these advantages make the FFHB electrolyte highly suitable for high-voltage applications.

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Interfacial Properties and Kinetic Analysis

To investigate the impact of Li^+ solvation structure on the electrochemical stability of the electrolyte, we first evaluated their basic physicochemical properties. As shown in Figure S10, FF, FFTB, and FFHB electrolytes exhibit viscosities of 6.25, 6.24, and 6.76 mPa·s, with corresponding ionic conductivities of 6.82, 4.86, and 4.16 mS cm^{-1} (Figure S10a). Although the optimized solvation structure in FFHB slightly decreases the ionic conductivity, it affords superior storage stability and improved oxidative stability compared with the FF electrolyte (Figure S10b,c). Linear sweep voltammetry (LSV) results further revealed that the onset oxidation potentials of FFHB and FFTB were over 0.2 V higher than that of FF (Figure S10b), indicating their enhanced oxidative stability under high-voltage conditions. As shown in Figure 3a, the electrochemical floating tests exhibited a lower leakage current density in FFHB electrolyte, which indicates that parasitic reactions are suppressed at the $LiCoO_2$ cathode interphase. This is further supported by the post-test SEM analysis. Specifically, following electrochemical floating tests, the $LiCoO_2$ cathode in the FFHB electrolyte exhibited a smoother surface with minimal cracking, in stark contrast to the severely degraded surfaces observed in the FF and FFTB electrolytes, implying that a well-defined CEI is formed at the surface of $LiCoO_2$. To further assess the degree of cathode degradation in different electrolytes, the transition metal dissolution analysis was conducted after 300 cycles at 0.2C. Transition metal dissolution analysis revealed that the amount of dissolved Co in FFHB was lower than that in FF and FFTB electrolytes (Figure 3b), confirming its superior capability in mitigating cathode degradation under high-voltage. Figures S11, S12a–c provides additional evidence of the impact of electrolyte formulation on the structural integrity of $LiCoO_2$ electrodes. After 300 cycles in FFHB, $LiCoO_2$ particle maintained an intact structure and a relatively smooth surface. In contrast, the samples cycled in the FF electrolyte suffered severe fragmentation. Furthermore, the structural degradation of the $LiCoO_2$ cathode can be further detected by ex-situ X-ray diffraction (XRD) and scanning transmission electron microscope (STEM). The $I_{(003)}/I_{(104)}$ peak intensity ratio of the $LiCoO_2$ electrode cycled in the FFHB electrolyte (8.57) was higher than that in FF and FFTB (7.57 and 8.39, respectively, Figure S13), suggesting better structural integrity.[50,51] High-resolution transmission electron microscopy (HRTEM) and corresponding fast Fourier transform (FFT) patterns revealed that the $LiCoO_2$ electrode cycled in FFHB exhibited only sporadically distributed spinel phase regions on its surface (Figure S12d–f), covering a much smaller area compared with those observed in FF and FFTB electrolytes.[1,10,52] These structural findings are consistent with electrochemical analysis. Ex-situ electrochemical impedance spectroscopy (EIS) demonstrated significantly lower interfacial impedance for FFHB, implying suppressed side reactions during long-term cycling (Figures 3c, S14). Moreover, differential capacity (dQ/dV) curves further confirmed this stability (Figure S15), showing minimal polarization and well-preserved redox features for the FFHB-cycled electrode compared to FF and FFTB. Together, these results indicate that the FFHB electrolyte effectively mitigates parasitic reactions and inhibits the degradation of $LiCoO_2$ under high-voltage cycling.

To gain deeper insight into the chemical composition and spatial distribution of CEI on $LiCoO_2$ cathode in the three electrolytes, in-depth X-ray photoelectron spectroscopy (XPS) was employed. The C 1 s spectra were deconvoluted into four components, including the C─C/C─H peak (284.8 eV), the C─O peak (286.1 eV), the C═O peak (287.9 eV), and the C─F peak (290.2 eV). Among them, the C─O and C═O signals originate from organic decomposition products of carbonate solvents and represent unstable CEI species under high-voltage conditions. As shown in Figure S16, these peaks became more pronounced in the FF electrolyte and were significantly reduced in the FFHB electrolyte with increasing sputtering time (0–320 s), indicating suppressed solvent decomposition in FFHB. In the O 1 s and F 1 s spectra (Figures 3d,e, S17, S18), the FFHB-derived CEI exhibits significantly stronger Li_2O (529.2 eV) and LiF (685.7 eV) signals compared to those in FF and FFTB, regardless of sputtering depth. This indicates a stable and abundant presence of Li_2O and LiF throughout the CEI, supporting the formation of a robust, inorganic-rich interphase. The differences in interfacial composition among these three electrolytes can be fundamentally attributed to their distinct Li^+ solvation structures. In particular, the FFHB electrolyte, characterized by compact and anion-enriched solvation sheath, promotes the preferential formation of inorganic-rich CEI components such as LiF and Li_2O. This inference is further supported by the B 1 s spectra (Figure S19), where both B–F (193.7 eV) and B–O (191.7 eV) signals were detected in FFTB and FFHB. Notably, the FFHB-derived CEI retained B-containing species even after extended sputtering time, while the corresponding signals in FFTB diminished rapidly with depth. These results suggest that the FFHB-derived CEI features a bilayered architecture, comprising an outer layer enriched with organic moieties (C─O, C═O, B─F, and B─O) and an inner layer dominated by inorganic species such as Li_2O and LiF (Figure 3f). This compositional gradient originates from the unique solvation environment in FFHB, which promotes preferential anion-derived interphase formation. Such a CEI configuration not only enhances interfacial mechanical and chemical stability but also effectively suppresses continuous electrolyte decomposition and transition metal dissolution, enabling durable high-voltage operation.

The enriched presence of inorganic components (LiF/Li_2O) in the CEI is beneficial to enhance both ionic conductivity and interfacial stability, thereby facilitating improved charge transport at the electrode–electrolyte interphase.[53–60] To evaluate interfacial kinetics, in-situ EIS coupled with distribution of relaxation time (DRT) analysis was performed.[61–64] As shown in Figures 3g,h, S23, S24, the FFHB electrolyte exhibits the lowest interphase resistance (R_{CEI}) and charge transfer resistance (R_{ct}) during cycling, indicating faster Li^+ transport and a more stable CEI. Temperature-dependent EIS measurements combined

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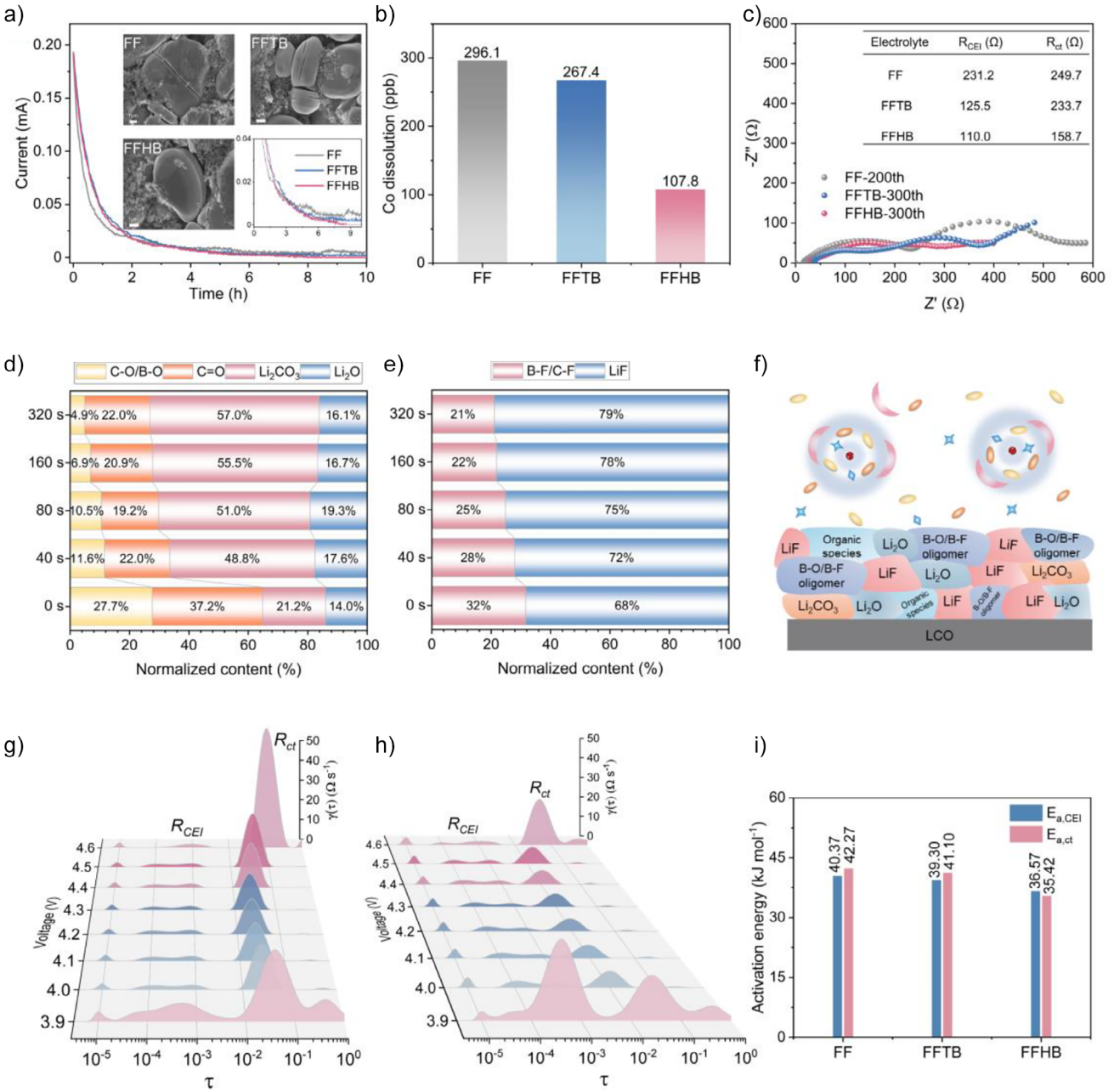


Figure 3. Interfacial chemistry and kinetic analysis of the $LiCoO_2$ cathode in different electrolytes. a) Leakage current during a constant-voltage (4.7 V versus Li^+/Li) floating test of the $LiCoO_2$ cathode after formation cycles in different electrolytes (inset: SEM image of $LiCoO_2$ cathode after the floating test). b) Transition metal dissolution determined by ICP-MS after 300 cycles at 0.2C. c) EIS spectra and corresponding fitting results of $LiCoO_2$//Li coin cells after cycling at 0.3C. d,e) Normalized ratios of O 1 s and F 1 s in the CEI formed from FFHB. f) Schematic diagram of the CEI in FFHB electrolyte. The DRT curves derived from the in-situ EIS of $LiCoO_2$//Li cells using g) FF and h) FFHB electrolytes. i) Activation energy of the resistance corresponding to Li^+ transport through CEI and charge-transfer processes.

with Arrhenius fitting further reveal that FFHB achieves the lowest activation energies for both Li^+ conduction across the CEI ($E_{a,CEI,}$ 36.57 kJ mol^{-1}) and charge transfer ($E_{a,ct}$, 35.42 kJ mol^{-1}), suggesting reduced energy barriers (Figures 3i, S20, S21). Additionally, galvanostatic intermittent titration technique (GITT) tests (Figure S22) confirm higher Li^+ diffusion coefficients and lower voltage hysteresis in the FFHB electrolyte, indicating improved lithium-ion mobility and reduced polarization during high-voltage cycling.

Electrochemical Performance Analysis

Following analysis of the composition and nanostructure of the formed CEI, the FFHB electrolyte was evaluated for its ability to stabilize $LiCoO_2$//Li coin cells under high-voltage conditions (cathode loading ~11 mg cm^{-2}). We first investigated the impact of varying HFE ratios on electrolyte properties and electrochemical performance. As shown in Figures S25–S28, excessive HFE content (≥40%) suppresses

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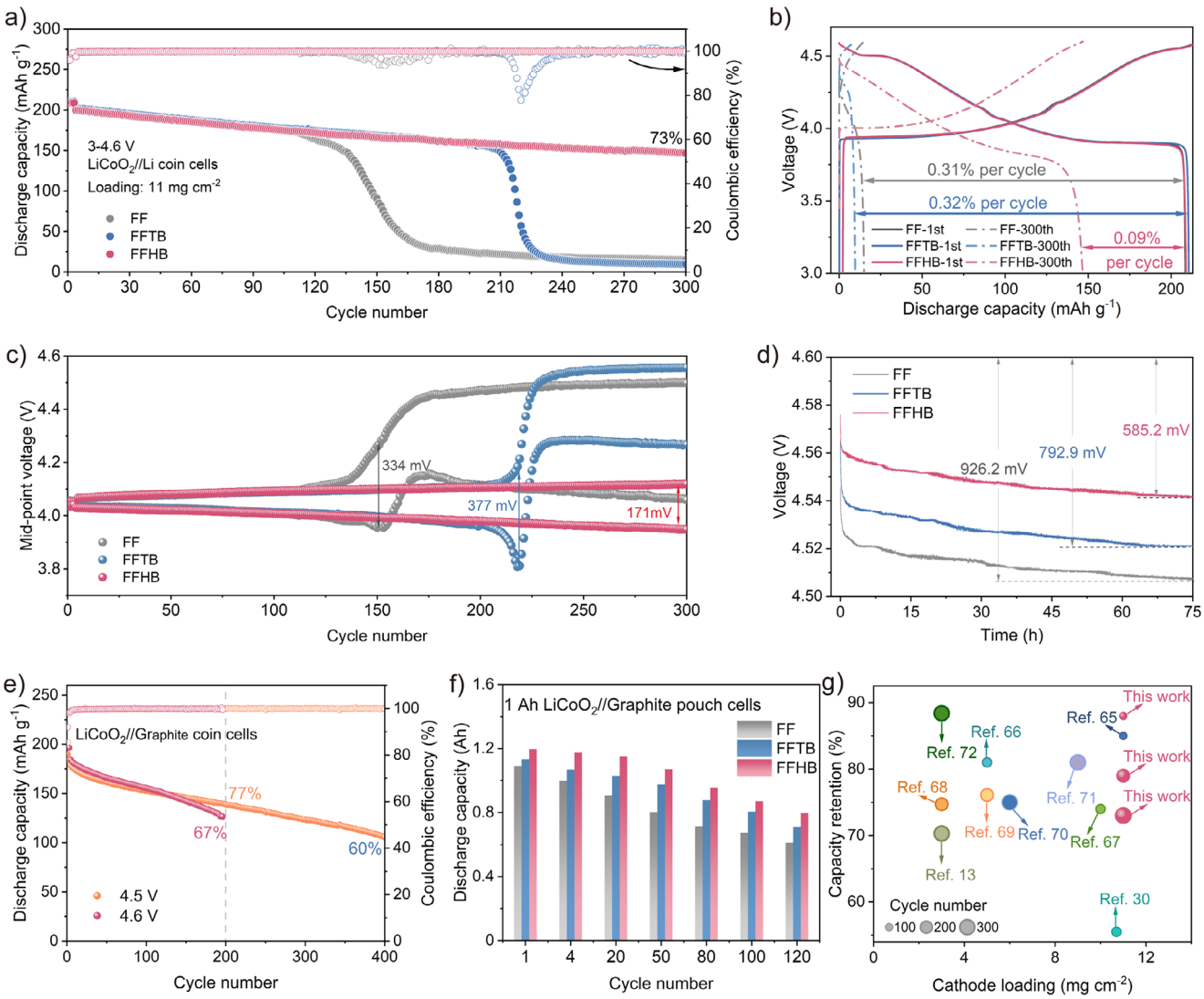


Figure 4. The electrochemical performance of studied electrolytes. a) Cycling performance of the $LiCoO_2$//Li coin cells with a cut-off voltage of 4.6 V, and b) corresponding charge–discharge profiles. c) Mid-point voltage of three electrolytes. d) Self-discharge test of $LiCoO_2$//Li coin cells. e) Cycling performance of the $LiCoO_2$//Graphite coin cells with FFHB electrolyte. f) Cycling performance of the $LiCoO_2$//Graphite pouch cells in different electrolytes at 4.5 V. g) Comparison of high-voltage cycling stability between $LiCoO_2$//Li coin cells in this work and previous results.

salt dissociation, lowers ionic conductivity, and results in poor cycling stability. In contrast, the 20% HFE electrolyte achieves the best balance of solvation chemistry, transport properties, and interfacial stability, delivering the highest Li^+ transference number, and capacity retention. These results clearly identify FFHB with 20% HFE as the optimal electrolyte formulation. As shown in Figure 4a, $LiCoO_2$//Li coin cells using FF and FFTB electrolytes exhibited rapid capacity degradation, resulting in failure after approximately 150 and 220 cycles at 0.2C, respectively. In contrast, the FFHB electrolyte enabled markedly improved cycling stability, achieving 73% capacity retention after 300 cycles, with an average capacity decay of only 0.21 mAh g^{-1} per cycle (Figure 4b), highlighting the mitigated irreversible phase transition and structural degradation in FFHB electrolyte. Further analysis of voltage polarization provides additional evidence for the interfacial stability and fast Li^+ kinetics enabled by the FFHB electrolyte. Notably, a sharp increase in voltage polarization at approximately the 150th and 220th cycle was observed in $LiCoO_2$//Li coin cells with FF and FFTB electrolytes, respectively, indicating accelerated interfacial degradation and failure of the $LiCoO_2$ cathode. However, the cells employing FFHB electrolyte maintained a much lower and more stable polarization throughout the cycling period (Figure 4c), indicating the formation of a robust CEI and efficient charge transport across the interphase. Additionally, the self-discharge behavior of these three electrolytes was evaluated, as shown in Figure 4d. $LiCoO_2$//Li coin cells using FF and FFTB electrolytes exhibited a rapid voltage drop after being rested for 75 h, indicating severe parasitic reactions and poor charge retention, whereas cells with FFHB electrolyte maintained a much higher voltage, suggesting suppressed side reactions under high-voltage conditions. Building upon the promising performance in $LiCoO_2$//Li

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half cells, the FFHB electrolyte was further evaluated in full-cell systems. The assembled $LiCoO_2$//Graphite coin cells exhibited excellent cycling stability under 4.5 and 4.6 V, confirming the compatibility of FFHB with both high-voltage cathodes and graphite anodes (Figure 4e). Beyond coin cells, the cycling performance of $LiCoO_2$//Graphite pouch cells was tested at a cut-off voltage of 4.5 V (Figure 4f). Notably, the $LiCoO_2$//Graphite pouch cells with FFHB electrolyte exhibited excellent cycling stability (retaining 0.71 Ah capacity after 150 cycles), outperforming its counterparts using FF and FFTB electrolytes (Figure S29). These results demonstrate that the modified electrolyte enables long-term cycling durability in both high-loading $LiCoO_2$//Li coin-type and commercial $LiCoO_2$//Graphite pouch cells, showing competitive performance compared to recently reported high-voltage lithium-metal battery systems (Figure 4g and Table S1).[13,30,65–75]

Conclusion

In summary, we have tailored the solvation structure of fluorinated electrolytes by modulating intermolecular interactions through a synergistic combination of the inductive effect and steric hindrance effect, thereby enhancing the high-voltage stability of $LiCoO_2$-based batteries. The designed diluent, 1*H*,1*H*,5*H*-perfluoropentyl-1,1,2,2-tetrafluoroethylether (HFE), featuring high steric hindrance and strong electron-withdrawing inductive effects, significantly weakens Li^+–solvent interactions, increases the proportion of CIPs in the solvation structure, and facilitates anions coordination. This optimized solvation environment enables the formation of a robust CEI featuring an inorganic-rich inner layer (offering high stability and ionic conductivity) and an organic-rich outer layer (providing high mechanical flexibility). As a result, the HFE-containing FFHB electrolyte exhibits excellent electrochemical stability under high-voltage conditions, enabling high-loading $LiCoO_2$//Li cells to stably cycle for over 300 cycles at 4.6 V and achieving 77% capacity retention for over 200 cycles in $LiCoO_2$//Graphite full-cells at 4.5 V. This work not only advances the development of electrolytes for high-voltage $LiCoO_2$-based batteries but also provides a valuable framework for rational electrolyte design.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

All data are available in the main manuscript or the Supporting Information.

Keywords: Electrolyte · Fluorinated solvent · $LiCoO_2$ · Lithium-ion battery

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